TRM4.v3 — Technical Research Review and Recommended Improvements

Physical Modeling, Detector Characterization, Radiometry, MTF, and DRI Performance Assessment

Independent technical review | Companion to Fraunhofer IOSB Technical Report 2021/04
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Reference document: Fraunhofer IOSB, *TRM4.v3 — Parameter Description and Model Documentation*, Technical Report 2021/04, 216 pages [R0]
Reference sources: EMVA 1288; ISO 12233:2024; ISO 15739:2023; ISO 9335:2025; HITRAN2024 (2026 publication); JCGM/GUM; sensor manufacturers' technical literature [R1–R14]

Purpose and scope. This report consolidates an actionable set of improvements for using TRM4.v3 in visible, NIR/SWIR, and thermal imaging applications. It distinguishes source-supported findings, engineering interpretation, and proposed extensions. It is **not** a corrigendum issued by Fraunhofer IOSB and does not claim to have formally verified every algebraic equation in the original 216-page technical report. All major document chapters and appendices are represented in the coverage map; recommendations address the principal model assumptions and input interfaces reviewed in the preceding research.

Executive Summary

Principal conclusion. There is no demonstrated need to replace the physical foundations of TRM4.v3 wholesale. Radiometry, photon-to-electron conversion, shot-noise physics, sampling theory, motion-induced MTF effects, and the Johnson/AMOP framework remain meaningful within their stated assumptions. The most consequential risks arise when generic model inputs do not correspond to the *actual operating point* of a camera: detector conversion-gain setting, readout mode, saturation node, spectral response, and image-processing state [R0–R7].

Three implementation issues have **critical** priority. First, distinguish intrinsic full-well capacity from camera-level saturation and ADC clipping. Second, specify the precise QE convention and its corresponding pixel area; otherwise fill factor may be double-counted. Third, prevent duplicate application of diffraction or aberration terms when importing a complete optical MTF from Zemax or a test laboratory.

More recent reference material should inform measurement protocols and databases, not be misrepresented as changing immutable physics. The previous review identified ISO 12233:2024 for digital-camera SFR, ISO 15739:2023 for digital-image noise, ISO 9335:2025 for optical transfer-function measurements, and the 2026 publication concerning HITRAN2024 molecular spectroscopy [R3–R6]. These apply within their respective scopes; extension to LWIR and uncooled bolometers requires dedicated validation.

Recommended implementation order: establish a traceable sensor-characterization layer; separate measured/manufacturer/assumed inputs; add automatic consistency and unit checks; import calibrated optics and atmosphere data; then extend sensitivity, uncertainty, temporal processing, and observer/DRI validation.

1. Methodology, Evidence Types, and Limitations

The earlier review compared the uploaded Fraunhofer report [R0] against standards publishers (ISO, EMVA, JCGM), relevant peer-reviewed literature, the HITRAN database, and manufacturer documentation. Evidence priority is (1) applicable primary standards and publications, (2) peer-reviewed technical studies, (3) device-specific manufacturer measurements, and (4) engineering inference. Where a cited standard is paywalled, the available public edition metadata and scope—not inaccessible normative text—support claims about its edition and applicability.

Classification	Interpretation in this report
Required correction	A definition or implementation may produce misleading or physically inconsistent output; clarify, validate, or warn.
Newer reference	A post-2021 standard or publication supports updating procedures, data libraries, or traceability.
Recommended extension	An additional feature is useful but is not evidence that the original equations are wrong.
Retain / no change	Preserve the established physics and make its assumptions explicit.
Laboratory validation	A proposed test or calibration step rather than a normative mathematical amendment.

Limitations. Manufacturer data sheets characterize particular devices, not universal detector behavior. Standards for digital still-picture cameras cannot be applied automatically as fully validated LWIR/bolometer measurement procedures. Quantitative corrections to actual DRI ranges require the specific TRM input set, camera operating conditions, calibrated data, and field/observer validation. Johnson criteria are task-dependent perceptual constructs; they must not be presented as universal automatic-recognition probabilities.

2. Coverage Map of the Original TRM4.v3 Documentation

Original sections / pages	Scope in TRM4.v3	Recommended disposition

Original sections / pages	Scope in TRM4.v3	Recommended disposition
1.1–1.3, pp. 13–16	TRM concept, validity, nominal range	Retain; supplement with uncertainty and applicability statement [R0, R9].
2.2–2.5, pp. 18–28	Imaging types, FOV, scan, microscan, binning	Extend operating-mode and readout timing metadata [R0, R1].
2.6, pp. 29–33	Optics, F-number, diffraction, aberration, external MTF	Validate measured/spectral MTF, avoid duplicate filtering [R0, R5].
2.7–2.9, pp. 33–44	Image intensifiers, coupling, geometry, CCD	Update readout and detector-technology profiles; clarify smear mechanisms [R0, R7].
2.10–2.11, pp. 45–55	Cooled/uncooled thermal detection and NETD	Operating-point traceability and thermal test conditions [R0].
2.12–2.13, pp. 55–65	Visible/general detectors, QE, dark current, well fill, noise	Differentiate QE conventions and camera-level saturation [R1, R2].
2.14–2.17, pp. 65–73	Analog/digital processing, ADC, video	Add content-dependent/nonlinear processing profiles [R2, R4].
2.18–2.23, pp. 73–83	Display, human observer, stabilization, prefilter data	Traceable SFR/OTF and observer measurements [R3, R5].
2.24–2.26, pp. 83–96	Target, background, scene, illumination	Spectral reflectance/BRDF, partial-pixel targets, scene variability [R0].
2.27–2.28, pp. 96–102	Atmosphere and turbulence	Versioned spectral RT inputs and scenario-specific C_n^2 [R6].
3.2–3.4, pp. 104–114	Geometry, sampling, optics	Validate angular units, spatial frequency, Nyquist [R3, R5].
3.5–3.8, pp. 114–134	Spectral sensitivity, photon/carrier flux, signals, well fill	Enforce unit and QE/capacity consistency [R0, R1].
3.9–3.10, pp. 134–142	Temporal/spatial noise, NETD, IETD	Maintain distinct temporal and spatial measurements [R1, R4].
3.11–3.12, pp. 142–158	AMOP, MTDP, MDSP, MCP, DRI	Keep observer framework; calibrate scenario-specific outcomes [R0].
3.13, pp. 158–160	NIIRS and GSD	Report separately from DRI and algorithmic classification [R0].
4.1–4.3, pp. 161–179	Component, system and imported MTFs	Improve provenance and import checks [R3, R5].
Appendix A, pp. 181–184	Scaling factors	Retain; expand automated dimensional validation [R0].
Appendix B, pp. 185–190	Diffraction-limited MTF and pupil geometry	Retain; state whether imported optics MTF is partial or complete [R5].
Appendix C, pp. 191–204	Preset QE, spectra, reflectance, irradiation, turbulence	Version and qualify presets; prefer measured device-specific data [R6–R8].
Appendix D, pp. 205 onward	Symbols, nomenclature, MTF notation	Add a measurement-node and operating-mode data dictionary [R0].

3. Detailed Findings and Recommended Changes

3.1 Camera Architecture and Operating Modes

Original sections: 2.2–2.5, 2.9, 3.2. **Priority: High. Sources:** [R0, R1, R7].

Assessment. TRM4 appropriately separates scanning, staring, interlace, microscan, and binning. Real sensor modules may also permit overlapping integration and readout, rolling/global shutter, ROI windows, and operating-mode-dependent gain and noise.

Recommended revision. Introduce an explicit camera *Operating Mode Profile*: shutter type, readout, triggering, ROI, hardware versus digital binning, frame rate, minimum/maximum exposure, permissible overlap, and environmental constraints. Never derive a sensor's maximum physical exposure solely from $1/\mathrm{FPS}$ without verifying its timing architecture.

Expected benefit. Prevents physically impossible exposure settings and inappropriate cross-mode SNR comparisons.

3.2 Optics, Entrance Pupil, and F-number

Original sections: 2.6, 3.4, Appendix B. **Priority: High. Sources:** [R0, R5].

Assessment. The paraxial on-axis irradiance factor $\pi/(4F^2)$ is appropriate for a simple filled-pupil, extended-source approximation. Fast systems, off-axis field points, obscurations, and vignetting require more complete pupil/throughput treatment.

Recommended revision. Document entrance-pupil geometry, effective F-number, etendue/numerical-aperture convention, spectral transmission as a function of field, and vignetting. Distinguish specified geometrical F-number from effective or working F-number, and use measured throughput when available.

Expected benefit. More accurate photon-rate and saturation predictions, especially toward field edges.

3.3 Imported Optical MTF and Zemax

Original sections: 2.6.5–2.6.8, 4.2.1, 4.3. **Priority: Critical. Sources:** [R0, R5].

Assessment. TRM4 explicitly states that externally supplied optical MTF replaces its internal optics-MTF calculation. If an imported MTF already includes diffraction, aberration, and focus, those components must not be multiplied into the transfer function again.

Recommended revision. Require frequency units (cycles/mm or cycles/mrad), sagittal/tangential axis, field position, wavelength/band, focus, temperature, source (modeled/measured), normalization, and an explicit **diffraction included?** flag. Validate axis orientation and frequency conversion. Make the internal and imported optics-MTF paths mutually exclusive at the corresponding modeling level.

Expected benefit. Prevents spuriously pessimistic or optimistic DRI predictions.

3.4 Detector Technologies and Spectral Families

Original sections: 2.3, 2.10–2.13, Appendix C.1. **Priority: High. Sources:** [R0, R8].

Assessment. Thermal, Visible/NIR/SWIR, and General remain useful TRM categories, but material name alone does not determine spectral cutoff, cooling needs, readout limits, or noise.

Recommended revision. Add versioned detector profiles for conventional/extended InGaAs, visible CMOS, cooled IR FPAs, and uncooled bolometers, with the applicable QE or spectral response, dark current or NETD, gain, and readout metadata. Never apply a photoelectron full-well model unmodified to a microbolometer.

Expected benefit. Correct physical submodel selection and defensible cross-technology comparisons.

3.5 Quantum Efficiency and Pixel Area

Original sections: 2.9.1–2.9.2, 2.10.7, 3.5, 3.7. **Priority: Critical. Sources:** [R0, R1].

Assessment. TRM's detector QE is expressed for its active detector area. Characterization reports may instead use a total/external QE convention referenced to the nominal full-pixel area, thereby accounting for microlenses and fill factor in the measured result.

Recommended revision. Require a `QE_basis` field (`intrinsic_active_area` or `total_pixel_area`). Store pitch and active dimensions independently. Perform a consistency check so that fill factor, microlens gain, or a geometric area correction is not applied twice.

Expected benefit. Eliminates a systematic bias in photoelectrons, SNR, integration time, and DRI.

3.6 Full-Well Capacity, Storage Capacity, and Operational Saturation

Original sections: 2.10.5, 2.12.8, 2.13.5, 3.8.2. **Priority: Critical. Sources:** [R1, R7].

Assessment. Physical pixel full well is not necessarily the limiting camera saturation capacity. The ROIC node, analog-chain gain, or ADC can clip earlier. Electron gain does not by itself redefine the upstream physical pixel well, but can alter the effective input-referred clipping threshold.

Recommended revision. Separate `Physical_FWC_e`, `Camera_Saturation_Capacity_e`, and `ADC_Clip_DN`; label each value's measurement node and acquisition mode. If the input-referred saturation capacity is unknown, issue a well-fill warning instead of treating a generic FWC as verified.

Expected benefit. Reduces unanticipated clipping and false confidence in long integration times.

3.7 Well Fill and Gain Modes

Original sections: 2.12.7–2.12.8, 3.5.7, 3.8.2. **Priority: Critical. Sources:** [R0, R1, R7].

Assessment. TRM's storage-gate equations are internally meaningful for the specified accumulation node; practical usable capacity, read noise, and conversion gain can depend on sensor mode.

Recommended revision. Associate saturation capacity, read noise, conversion gain, and dark current with the same gain/temperature/readout profile. Evaluate well fill for both the **reference background** and a **brightest expected pixel** scenario. Treat 70% fill as an illustrative design margin, not a universal safety standard.

Expected benefit. More reliable automated exposure selection and scenario range reports.

3.8 Integration Duty Cycle and Frame Rate

Original sections: 2.5.2, 2.12.7, 3.5.7. **Priority: High. Sources:** [R0, R7].

Assessment. The relationship between exposure, available integration time, and FPS depends on sensor timing. The shorthand $t_{\mathrm{int}} = \eta/\mathrm{FPS}$ only represents a simplified non-overlapping frame scheme.

Recommended revision. Provide a per-sensor exposure/readout timing diagram and explicitly compute the hardware timing limit, guard time, frame trigger and rolling/global shutter constraints. Report frame-rate limits separately from saturation and motion-blur limits.

Expected benefit. Avoids unrealistic photocharge and SNR predictions.

3.9 Radiometry and Per-Pixel Photon Rate

Original sections: 2.24–2.27, 3.5–3.8. **Priority: High. Sources:** [R0, R1, R6].

Assessment. The chain of spectral scene radiance → focal-plane spectral irradiance → photons → photoelectrons remains valid. Common errors involve confusing photon *rate* with count per exposure, mixing wavelength units, or treating an underfilled target as an extended uniform surface.

Recommended revision. Record spectral source and background signatures, atmospheric transmission and path radiance, wavelength/field-dependent optics throughput, QE convention, pixel sensitive area, and dark current. Provide automated unit checking and PSF/target-occupancy logic.

Expected benefit. Eliminates order-of-magnitude errors and improves small-target simulations.

3.10 Noise Sources and SNR Bookkeeping

Original sections: 2.10–2.13, 3.8–3.10. **Priority: High. Sources:** [R1, R2, R4].

Assessment. Shot, dark, read and quantization noise do not fully describe a post-processed video system. Fixed-pattern noise, PRNU/DSNU, spatial and temporal correlations, and nonlinear mappings must be classified separately.

Recommended revision. Characterize appropriate linear and nonlinear operating modes using the relevant EMVA 1288 approaches; label each noise value's measurement port (`RAW`, post-NUC, processed video). Within the ISO standard's scope, reference ISO 15739:2023 for digital-camera noise and related characteristics.

Expected benefit. Meaningful noise/SNR assessment across HDR and denoising configurations.

3.11 Dark Current as a Function of Temperature

Original sections: 2.12.6, 3.7.5. **Priority: Medium–High. Sources:** [R1, R8].

Assessment. A single dark-current value may not represent behavior across the full sensor-temperature and gain range. Dark current, hot pixels, and spatial variation require qualification by condition.

Recommended revision. Support a calibrated `I_dark(T_sensor, gain, mode)` table, include measurement spread, and extract values from dark RAW sequences. Do not impose the same temperature-doubling rule on all detector families.

Expected benefit. More credible low-light sensitivity predictions.

3.12 Uncooled Bolometers, NETD, IETD, and NUC

Original sections: 2.11, 3.9–3.10. **Priority: High. Sources:** [R0, R4] (ISO applicability must be checked).

Assessment. Bolometer response is thermal rather than the photocharge-well response used for photon detectors. Thermal time constant, readout/filtering, NETD, temporal noise, spatial fixed pattern, and NUC state must be independently represented.

Recommended revision. Define a blackbody measurement protocol: temperature steps, reference background, F-number, focus, frame rate, NUC state, number of frames, temporal standard deviation, spatial nonuniformity, `DN/K` response, and measurement port. Avoid equating thermal response time with exposure time.

Expected benefit. Reproducible LWIR comparisons and reliable entry of NETD/IETD into TRM.

3.13 Sampling, Aliasing, and Camera SFR

Original sections: 2.5.3, 3.3, 4.2.3, 4.2.6. **Priority: High. Sources:** [R0, R3].

Assessment. Nyquist and aliasing remain unchanged physical/sampling concepts. Resolving apparent high-frequency bars does not necessarily prove true non-aliased resolution.

Recommended revision. Adopt a traceable camera SFR measurement procedure, using the applicable methods and scope of ISO 12233:2024; document preprocessing, Nyquist frequency, overshoot, and aliasing. Do not equate an optical MTF with a processed-camera SFR.

Expected benefit. More reproducible resolution estimates.

3.14 Digital Processing, Video Chain, and Display

Original sections: 2.14–2.18, 4.2.4–4.2.9. **Priority: High. Sources:** [R0, R2, R3, R4].

Assessment. Linear MTF multiplication is insufficient for content-dependent sharpening, temporal denoising, local-contrast enhancement, HDR, gamma, and learned reconstruction.

Recommended revision. Map the processing stages from sensor RAW through NUC/demosaic/enhancement, compression, and display. Measure processed-output SFR/noise for each profile instead of multiplying unqualified MTFs for nonlinear algorithms.

Expected benefit. Links physical sensor predictions to what an operator actually sees.

3.15 Human Observer and Task Performance

Original sections: 2.18, 3.11–3.12. **Priority: Medium–High. Sources:** [R0, R4].

Assessment. TRM's AMOP/MCP/MTDP framework includes perceptual assumptions. Human performance depends on the operator, display, observation time, clutter, task definition, and psychometric threshold.

Recommended revision. Add structured observer trials, ground truth, success probabilities, confidence intervals, and explicit task descriptions. Do not treat AI classification as equivalent to human detection or identification.

Expected benefit. Empirical linkage of predicted DRI and actual operator success.

3.16 Motion Blur, Stabilization, and Smear

Original sections: 2.9.5, 2.19, 4.2.10. **Priority: High. Sources:** [R0, R5].

Assessment. Frame-transfer CCD smear and image-motion blur are different mechanisms, described separately in TRM. Vibration and relative image velocity can blur an exposure even without frame-transfer smear.

Recommended revision. Adopt unambiguous naming; calculate an exposure-motion budget from angular velocity, integration time, and IFOV; support measured vibration trajectories and PSF/MTF where assumptions of Gaussian random motion or constant velocity fail.

Expected benefit. Reduces overly optimistic task estimates under LOS disturbance.

3.17 Target, Background, and Illumination

Original sections: 2.24–2.26, 3.6. **Priority: Medium. Sources:** [R0, R6].

Assessment. Scalar reflectance and Lambertian targets are useful simplifications, but real materials can exhibit wavelength and viewing-angle dependence.

Recommended revision. Import spectral reflectance and emissivity, optional BRDF where calibrated, illumination geometry, shadows, target occupancy/PSF, and clutter descriptors. Do not infer an exact field spectrum from visible-range luminance alone.

Expected benefit. More realistic scene signal and contrast predictions.

3.18 Spectral Atmosphere and HITRAN2024

Original sections: 2.27, 3.7, Appendix C. **Priority: High. Sources:** [R0, R6, R13].

Assessment. TRM already allows external atmospheric transmission data; newer molecular spectroscopy provides grounds for updating the underlying spectral database and metadata, not for rejecting Beer–Lambert attenuation as a limiting approximation.

Recommended revision. Record spectroscopy database edition, radiative-transfer solver/version, gas concentrations, temperature-pressure-height profile, aerosol assumptions, spectral resolution, and output bandpass. Integrate transmission with QE/filter spectra rather than applying a single wavelength-independent loss where spectral structure matters.

Expected benefit. Better traceability in long-range and narrow-band SWIR/MWIR scenarios.

3.19 Path Radiance and Aerosol Scattering

Original sections: 2.27.4, 3.7.3. **Priority: High. Sources:** [R0, R6].

Assessment. A simplified path-radiance factor proportional to $(1-\tau)$ is convenient but does not resolve solar geometry, aerosol phase function, multiple scattering, or thermal atmospheric emission for all scenarios.

Recommended revision. Permit imported, versioned $L_{\mathrm{path}}(\lambda, R)$ from an independently validated radiative-transfer calculation. Keep transmitted target radiance and added path radiance as distinct terms with consistent spectral units.

Expected benefit. Less biased target-background contrast at extended ranges.

3.20 Turbulence and the C_n^2 Profile

Original sections: 2.28, 4.2.13, Appendix C.7. **Priority: Medium. Sources:** [R0, R10].

Assessment. Kolmogorov turbulence and Fried parameter r_0 remain useful baseline models, but simplified near-ground profiles need not represent slant or strongly time-dependent paths.

Recommended revision. Support measured or prescribed $C_n^2(s,t)$, altitude profiles, short/long-exposure MTF choices, and an optional von Kármán outer-scale model with explicit assumptions. Distinguish turbulence-derived image motion from residual platform jitter.

Expected benefit. Improved atmospheric MTF and honest confidence bounds.

3.21 OTF/MTF Measurement and Reconstruction

Original sections: 2.6, 2.20–2.23, 4.1–4.3. **Priority: High. Sources:** [R0, R5].

Assessment. ISO 9335:2025 is a newer optical-transfer-function measurement reference, potentially useful for updating acquisition, correction, and uncertainty procedures. It does not change the basic diffraction equations.

Recommended revision. Require focus, field point, sagittal/tangential directions, pupil, wavelength band, environment, calibration, measured transfer range, and uncertainty with every OTF/MTF file. Identify whether data are optical-only or end-to-end.

Expected benefit. Traceable comparisons among design prediction, bench measurements, and field evidence.

3.22 MRTD, MTDP, MCP, MDSP, and Johnson Criteria

Original sections: 2.21–2.23, 3.11–3.12. **Priority: High. Sources:** [R0, R4].

Assessment. The perceptual framework remains useful when observer and task assumptions are retained. Cycle count alone cannot describe all real cluttered scenes; a single CNR number alone is not a validated substitute for the Johnson/AMOP framework.

Recommended revision. Attach target dimensions, detection/recognition/identification task, cycle criterion, spatial contrast/MTF, observer protocol, measured success probability, and validity domain to every DRI result.

Expected benefit. Interpretable and auditable range claims.

3.23 NIIRS and Image Interpretability

Original sections: 3.13. **Priority: Medium. Sources:** [R0, R3].

Assessment. NIIRS estimates image interpretability in a separate assessment framework; its numerical output must not be merged directly with Johnson DRI or algorithmic confidence scores.

Recommended revision. Display NIIRS separately with associated GSD, edge response, scenario, and validation metadata. Map scales to one another only when supported by empirical calibration.

Expected benefit. Prevents metric conflation and misleading operational conclusions.

3.24 Preset Data and Parameter Lifecycle

Original sections: Appendix C. **Priority: High. Sources:** [R0, R7, R8].

Assessment. Preset QE curves, radiation sources, material signatures, and turbulence levels provide convenient nominal assumptions, not certified data for every contemporary detector.

Recommended revision. Record source, date, version, applicability, uncertainty, and usage warning for each preset; give priority to a correctly calibrated sensor-specific curve and preserve history when reference data change.

Expected benefit. Avoids obsolete or out-of-domain inputs silently affecting results.

3.25 Uncertainty, Sensitivity, and Monte Carlo Analysis

Original sections: 1.3, Chapter 3, Appendix D. **Priority: High. Sources:** [R9–R12].

Assessment. Nominal predictions depend on uncertain and sometimes correlated inputs, including QE, atmosphere, C_n^2, focus, vibration, and observer thresholds.

Recommended revision. Define an uncertainty budget and variance/Monte Carlo propagation with input distributions and correlations. Report scenario-specific percentiles ($P_{10}/P_{50}/P_{90}$) with the associated task performance metric and model validity range.

Expected benefit. Engineering decisions based on a documented uncertainty range rather than one deterministic distance.

3.26 Model Traceability and Experimental Validation

Original sections: 1.3, Chapters 3–4. **Priority: High. Sources:** [R0, R1, R3–R5].

Assessment. Dimensional consistency alone does not verify an imaging prediction. Validation requires a traceable bridge from radiometric input to RAW response, noise, MTF/SFR, observer data and DRI.

Recommended revision. Maintain a validation matrix with calibration records, acquisition settings, firmware/processing state, ground truth, model version, run identifiers, uncertainties, and comparisons over several ranges and conditions.

Expected benefit. Reproducibility, explainable model errors, and controlled performance claims.

4. Improved Mathematical Presentation — Signal, Sensitivity, and Exposure

This section is an **engineering validation layer** built around established detector physics and the notation of TRM's radiometric/carrier-flux sections; it is not a wholesale replacement of Fraunhofer's published equations. The photoelectron storage model below is appropriate for integrating photon detectors and **must not be copied unchanged into an uncooled microbolometer model**.

4.1 Radiance to Photon Flux: Define Each Physical Quantity

For a uniform extended object over one sensor pixel, separate *scene radiance transmitted along the path* from *radiance added by the path*:

$$L_{\lambda,\mathrm{sensor}}(\lambda, R) = \tau_{\mathrm{atm}}(\lambda, R)\, L_{\lambda,\mathrm{scene}}(\lambda) + L_{\lambda,\mathrm{path}}(\lambda, R).$$

Here L_λ denotes spectral radiance in W·m^{-2}·sr^{-1}·µm^{-1}; τ_{atm} is dimensionless; R is target range. This simplified expression assumes that the scene source term has already been defined correctly for emission/reflection and that all terms use consistent spectral units.

For a uniformly filled optical pupil, near the optical axis, a simple paraxial expression for spectral focal-plane irradiance is

$$E_{\lambda,\text{FP}}(\lambda) \approx \frac{\pi}{4F^2}\,\tau_{\text{opt}}(\lambda)\,L_{\lambda,\text{sensor}}(\lambda,R), \qquad F \equiv f/D.$$

This is an approximation to angular integration of radiance over the optical acceptance cone. For fast lenses, finite conjugates, field-dependent throughput, or obscured pupils, evaluate the appropriate optical etendue / image-space numerical aperture and vignetting rather than treating $\pi/(4F^2)$ as exact.

Photon arrival rate at one pixel:

$$\boxed{\dot{N}_{\text{ph}} = A_{\text{ref}} \int_{\lambda_1}^{\lambda_2} E_{\lambda,\text{FP}}\,(\lambda)\,\frac{\lambda}{hc}\,\mathrm{d}\lambda}$$

Photoelectron generation rate at one pixel:

$$\boxed{\dot{N}_{e,\text{opt}} = A_{\text{ref}} \int_{\lambda_1}^{\lambda_2} E_{\lambda,\text{FP}}\,(\lambda)\eta_{\text{QE}}(\lambda)\,\frac{\lambda}{hc}\,\mathrm{d}\lambda}$$

Use **active sensitive area** for QE specified per active area, or **full pixel area** when QE has been defined as an effective/total QE referred to that area. Do not mix the two conventions. When spectral irradiance is tabulated per μm, integrate over μm but convert $\lambda/(hc)$ to meters per photon-energy calculation.

Symbol	Meaning	Units / note
$L_{\lambda,\text{scene}}$	Spectral target/background radiance	W/(m²·sr·µm)
$L_{\lambda,\text{path}}$	Path radiance added to LOS	Same radiance units
$E_{\lambda,\text{FP}}$	Focal-plane spectral irradiance	W/(m²·µm)
$F = f/D$	Optical F-number	Dimensionless; geometry-dependent
A_{ref}	QE-consistent pixel reference area	m²
$\eta_{\text{QE}}(\lambda)$	Quantum efficiency	Fraction in $[0,1]$
$\dot{N}_{\text{ph}}$	Incident photons per second per pixel	photons/s/pixel
$\dot{N}_{e,\text{opt}}$	Photoelectrons per second per pixel	electrons/s/pixel
t_{int}	Charge integration time	s

4.2 Electron Counts, Saturation, Well Fill, and SNR

The accumulated **input-referred** charge count for an integrating detector, before downstream electronic gain, is approximately

$$N_e(t_{\text{int}}) = t_{\text{int}}\left(\dot{N}_{e,\text{opt}} + \dot{N}_{e,\text{dark}}\right), \qquad \dot{N}_{e,\text{dark}} = \frac{I_{\text{dark}}}{q}.$$

The usable capacity is tied to a **specific accumulation or clipping node and operating mode**:

$$w = \frac{N_e(t_{\text{int}})}{N_{\text{cap,usable}}(g,m)}, \qquad w_{\%} = 100w.$$

Here w denotes the fractional well fill, $w_{\%}$ its percentage, and g and m denote the gain and readout mode. The formula should be tied to the correct physical storage node; it does not imply that analog amplification consumes a pixel's intrinsic full-well charge.

For a permitted fill fraction α_{fill}, and an estimate of the **brightest operational pixel**:

$$\boxed{t_{\text{sat,max}} = \frac{\alpha_{\text{fill}}\, N_{\text{cap,usable}}(g,m)}{\dot{N}_{e,\text{bright}} + \dot{N}_{e,\text{dark}}}}$$

For a linear raw output, an elementary input-referred shot-plus-read-noise estimate is

$$\text{SNR}_{\text{pixel}} \approx \frac{N_{e,\text{signal}}}{\sqrt{N_{e,\text{signal}} + N_{e,\text{dark}} + \sigma_{\text{read}}^2 + \sigma_{\text{other}}^2}}.$$

Interpretation. Specify whether $N_{e,\text{signal}}$ is total collected optical charge or target-minus-background contrast. For a difference between separate target and background estimates, noise variances, covariance, averaging, and processing must be handled explicitly; simply substituting a contrast numerator into a total-signal denominator can be wrong. Avoid double-counting dark, spatial, quantization or post-processing terms [R0–R2].

4.3 Timing and Motion-Blur Constraints

A defensible exposure recommendation solves several constraints jointly:

$$\boxed{t_{\text{int}} \le \min\left[t_{\text{sat,max}},\ t_{\text{timing,max}},\ t_{\text{motion,max}},\ t_{\text{hardware,max}}\right]}$$

For approximately constant angular image velocity ω_{rel}, pixel angular pitch (IFOV), and allowed smear B_{allow} in pixels,

$$B_{\text{smear}} = \frac{|\omega_{\text{rel}}|\, t_{\text{int}}}{\text{IFOV}}, \qquad t_{\text{motion,max}} = \frac{B_{\text{allow}}\, \text{IFOV}}{|\omega_{\text{rel}}|}.$$

Finally, verify that the selected time and gain yield the required task-specific SNR/contrast:

$$\text{SNR}(t_{\text{int}}, g, m) \ge \text{SNR}_{\text{required}}.$$

The timing term is sensor-specific. In a simple non-overlapping acquisition mode it may be at most $1/\text{FPS}$, but this should **not** be generalized to all sensor architectures.

4.4 Corrected SWIR Worked Example — Full Spectral Integration

Assume uniform source spectral radiance of $0.4\ \text{W}/(\text{m}^2\,\text{sr}\,\mu\text{m})$ over $1.2–1.7\ \mu\text{m}$, F/2 optics, combined optics-and-atmosphere transmission 0.80, pixel pitch $15\ \mu\text{m}$, total QE 0.80, and usable capacity 1.0×10^6 electrons. Assume full pixel coverage by the source, on-axis imaging, constant transmission/QE, no dark current, and no earlier clipping node.

Step A. Focal-plane spectral irradiance:

$$E_{\lambda,\text{FP}} \approx 0.4 \times 0.80 \times \frac{\pi}{4 \times 2^2} = 0.0628319\ \frac{\text{W}}{\text{m}^2\,\mu\text{m}}.$$

Step B. Spectral radiant power per pixel and integrated power:

$$A_{\text{pixel}} = (15 \times 10^{-6})^2 = 2.25 \times 10^{-10}\ \text{m}^2,$$

$$P_{\text{pixel}} = A_{\text{pixel}} E_{\lambda,\text{FP}}\,(1.7 - 1.2) = 7.06858 \times 10^{-12}\ \text{W}.$$

Step C. Spectrally integrated photon arrival rate. Because E_λ is constant across this example band, the energy-weighted mean wavelength for the integral is $1.45\ \mu\text{m}$, **not** $1.50\ \mu\text{m}$:

$$\dot{N}_{\text{ph}} = \frac{P_{\text{pixel}}\,(1.45 \times 10^{-6}\ \text{m})}{h\,c} = 5.15969 \times 10^{7}.$$

This result is a **per-pixel photon arrival rate** in photons/s. The corresponding number of photons collected during a frame depends on the integration time.

Step D. Electrons, exposure, and well fill:

$$\dot{N}_{e,\text{opt}} = 0.80\,\dot{N}_{\text{ph}} = 4.12775 \times 10^{7}\ \text{e}^{-}/\text{s},$$

$$N_e(10\ \text{ms}) = \dot{N}_{e,\text{opt}} \times 0.010 \approx 4.12775 \times 10^{5}\ \text{e}^{-},$$

$$w_{\%}(10\ \text{ms}) \approx 41.28\%, \qquad t_{70\%} \approx 16.96\ \text{ms}.$$

Parameter / result	Value
F-number; combined transmission	F/2; 0.80
Pixel area	$2.25 \times 10^{-10}\ \text{m}^2$
Spectral irradiance at focal plane	$0.0628319\ \text{W}/(\text{m}^2\ \mu\text{m})$
Integrated pixel optical power	$7.06858 \times 10^{-12}\ \text{W}$
Approximation using $\lambda = 1.50\ \mu\text{m}$	5.34×10^{7} photons/s
Spectrally integrated photon rate	5.160×10^{7} photons/s
Photoelectron generation rate	4.128×10^{7} electrons/s
Full-well fraction for 10 ms	41.28%
Time for illustrative 70% fill	16.96 ms

Interpretation. The monochromatic approximation at $1.50\ \mu\text{m}$ overestimates the photon rate by about 3.45% relative to the spectral integral for this deliberately flat radiance case. This is a refinement of the numerical illustration, **not** a correction to TRM4's spectral-integral formulation. A 70% target is an engineering choice, not a universal exposure safety limit.

5. Proposed Laboratory and Field Validation Plan

Before acceptance of software changes or formal range claims, establish test conditions, calibration evidence, and model-to-measurement comparison criteria.

Test ID	Activity	Key output	Reference framework
T01	RAW response vs. calibrated illumination	Conversion gain, response linearity, saturation, e-/DN	EMVA 1288 [R1, R2]
T02	Dark frames across temperature/exposure	$I_{\text{dark}}(T)$, DSNU, dark noise	EMVA 1288 [R1]
T03	Calibrated-radiance source	Radiance-to-RAW mapping and QE/area convention	TRM radiometric chain [R0, R1]
T04	Slanted-edge/sinewave acquisition	End-to-end camera SFR, overshoot, aliasing	ISO 12233:2024 within scope [R3]
T05	Calibrated lens bench	OTF/MTF by field, orientation, focus	ISO 9335:2025 [R5]

Test ID	Activity	Key output	Reference framework
T06	Blackbody and image sequences	NETD, NUC state, temporal/spatial pattern	Detector-specific thermal methods [R0]
T07	Vibration and angular-motion profiles	Motion PSF/MTF vs. exposure time	TRM stabilization model [R0]
T08	Field tests over multiple ranges	Contrast/SNR/MTF/task performance vs. atmosphere	TRM + atmospheric data [R0, R6]
T09	Input uncertainty propagation	DRI percentile intervals, model sensitivity	JCGM/GUM [R9–R12]

Mandatory test metadata: date, sensor and optics IDs, firmware, gain/readout mode, FPS, ROI, exposure, F-number, focus state, NUC/processing state, sensor temperature, target/background environment, raw-file references, calibration identifiers, traceable source data, and measurement uncertainty. Report processed-video SFR separately from lens-only optical MTF.

6. Implementation Roadmap and Acceptance Criteria

Phase	Changes	Suggested acceptance evidence
P0 — Correctness	QE convention and area; physical FWC vs. actual saturation; mode-aware well fill; prevent duplicate optics MTF	Every calculation reports gain/mode, capacity basis, QE basis and MTF provenance; inconsistent entries are rejected or flagged.
P1 — Calibration	Calibrated photon/dark/noise data; measured SFR/OTF; temperature curves; bright/dark well-fill cases	Bench measurements and predictions agree within a predefined measurement-uncertainty budget.
P2 — Advanced physics	Spectral atmosphere and path radiance; C_n^2 profiles; dynamic motion PSF	Documented comparisons at multiple distances, times and environmental conditions.
P3 — Observers and uncertainty	DRI uncertainty propagation; observer trials; nonlinear video processing/HDR	Probability-qualified ranges with assumptions, run ID, validation evidence and confidence statement.

7. Suggested Insert-Ready Wording for an Updated TRM Specification

7.1 Charge Storage and Saturation Capacity

“Specified storage capacity shall refer to the accumulation node represented by the adopted detector model. It shall not automatically be interpreted as the physical full well of an individual pixel. Operational saturation can occur at a ROIC node, analog amplifier, ADC, or later processing stage. The capacity definition, gain/readout mode, measurement node, and evidence source shall be reported.”

7.2 Quantum Efficiency Convention

“Each imported QE spectrum shall identify the wavelength range, data source, publication or measurement date, and whether the result refers to intrinsic active-area efficiency or a total pixel-area-referenced efficiency. Geometric fill factor and microlens effects shall not be counted twice.”

7.3 Well Fill and Exposure Selection

“Maximum usable integration time shall be assessed with a gain- and readout-mode-specific saturation capacity. Both reference background and brightest credible scene pixels shall be checked. The tool shall report infeasible combinations of saturation, motion, timing and SNR constraints.”

7.4 External Optical MTF

“Imported optics MTF data shall be accompanied by spatial-frequency units, field and orientation, wavelength/band, focus and operating temperature, and a declaration of whether diffraction, aberration and other

blur effects are already included. Transfer-function components contained within the imported data shall not be applied again."

7.5 Noise Characterization

"Noise measurements shall identify the camera operating mode and processing stage. Linear RAW, NUC-corrected, nonlinear HDR and displayed-video outputs shall be treated as distinct characterization conditions. Measurement standards shall be applied only within their validated scope."

7.6 Atmosphere

"For spectrally sensitive long-range scenarios, atmospheric transmission and path radiance should be available as functions of wavelength and range. The spectroscopy database, radiative-transfer software, atmospheric profile, spectral resolution and relevant assumptions shall be versioned and recorded."

7.7 DRI Reporting

"Every predicted DRI range shall be associated with task definition, target geometry, optical and detector operating point, illumination and atmosphere, MTF source, observer assumptions, uncertainty statement and validation status. A nominal range is not, by itself, an assured operational outcome."

8. References and Source Documents

The reference list below is carried forward from the previous Hebrew research report. Edition dates and linked locations are retained as originally documented; this English rewrite does not constitute a separate re-verification of every external publication or URL.

[R0] Fraunhofer IOSB (2021). *TRM4.v3 — Parameter Description and Model Documentation*. Technical Report 2021/04, supplied 216-page source document. Main primary source for TRM definitions, parameter locations and modeling assumptions.

[R1] EMVA (2021). *EMVA 1288, Release 4.0 Linear*. Linear image-sensor response, quantum efficiency, saturation, gain and noise characterization.
https://www.emva.org/wp-content/uploads/EMVA1288Linear_4.0Release.pdf

[R2] EMVA (2021). *EMVA 1288, Release 4.0 General*. Characterization framework for more general/nonlinear imaging systems.
https://www.emva.org/wp-content/uploads/EMVA1288General_4.0Release.pdf

[R3] ISO (2024). *ISO 12233:2024 — Digital cameras — Resolution and spatial frequency responses*. Fifth edition.
https://www.iso.org/standard/88626.html

[R4] ISO (2023). *ISO 15739:2023 — Photography — Electronic still-picture imaging — Noise measurements*. Fourth edition.
https://www.iso.org/standard/82233.html

[R5] ISO (2025). *ISO 9335:2025 — Optics and photonics — Optical transfer function — Principles and procedures of measurement*. Third edition.
https://www.iso.org/standard/85989.html

[R6] Gordon, I. E., et al. (2026). *The HITRAN2024 molecular spectroscopic database*. *Journal of Quantitative Spectroscopy and Radiative Transfer* 353, 109807. DOI: 10.1016/j.jqsrt.2026.109807.
https://doi.org/10.1016/j.jqsrt.2026.109807

[R7] Teledyne FLIR. *A6260 InGaAs Camera*, manufacturer data sheet (document issue date not independently verified). Device-specific illustration of capacity/gain and readout information.
https://flir.netx.net/file/asset/10276/original/attachment

[R8] Hamamatsu Photonics. *InGaAs area image sensors*, continuously updated manufacturer product catalogue; individual data-sheet revision dates vary.
https://www.hamamatsu.com/us/en/product/optical-sensors/image-sensor/ingaas-image-sensor/ingaas-area-image-sensor.html

[R9] JCGM (2026). *JCGM 100:2008/Amd.1:2026 — Nonlinearity in measurement models*.
https://doi.org/10.59161/PPDI3267

[R10] JCGM (2020). *JCGM GUM-6:2020 — Developing and using measurement models*.
https://doi.org/10.59161/JCGMGUM-6-2020

[R11] JCGM (2008). *JCGM 101:2008 — Propagation of distributions using a Monte Carlo method*.
https://doi.org/10.59161/JCGM101-2008

[R12] JCGM (2023). *JCGM GUM-1:2023 — Introduction*.
https://doi.org/10.59161/JCGMGUM-1-2023

[R13] HITRAN Online. Official citation and release policy.
https://hitran.org/citepolicy/

[R14] EMVA. Official EMVA 1288 download and release-history pages.
https://www.emva.org/standards-technology/emva-1288/emva-standard-1288-downloads-2/

9. Appendix — Recommended Input Dictionary for an Improved TRM Workflow

Information group	Required or preferred fields
Optics	Focal length; F-number; entrance pupil; field; imported MTF completeness; $\tau_{\mathrm{opt}}(\lambda)$; distortion; focus state
Detector	Technology; band; pitch; active dimensions; QE basis and $\eta_{\mathrm{QE}}(\lambda)$; dark current vs. temperature; read noise; PRNU/DSNU
Readout / gain	Gain and conversion-gain mode; physical/usable saturation; ADC clipping; HDR/RAW; FPS; ROI; shutter and timing
Scene	Target geometry; spectral reflectance/emissivity; background; source spectrum; range; solar/view geometry; target occupancy
Atmosphere	Temperature; RH; pressure; aerosol/visibility; vertical profile; $\tau_{\mathrm{atm}}(\lambda, R)$; $L_{\mathrm{path}}(\lambda, R)$; RT dataset version
MTF / motion	Optical vs. system MTF; tangential/sagittal axis; pixel aperture; Nyquist; LOS jitter/PSD; exposure trajectory; turbulence
Observer / task	D/R/I task; target criterion; display; viewing time; clutter; psychophysical validation status
Provenance	Physical units; source version/URL; publication date; measured/assumed status; uncertainty; run identifier; timestamp

10. Scientific Conclusions and Design Decision

Retain TRM4.v3's principal physical equations and focus immediate engineering changes on **input semantics, sensor-operating-mode consistency, calibrated characterization and traceability**. Three checks should be mandatory before an application issues DRI estimates: (1) QE and geometric pixel-area definitions must be mutually consistent; (2) the saturation capacity used for well-fill calculations must be valid for the modeled readout node and gain state; and (3) an imported optical MTF must be applied once, with its content and units documented.

Subsequent improvements should prioritize measured end-to-end SFR/OTF, wavelength-dependent atmospheric transmission and path radiance, realistic processing-pipeline behavior, and probabilistic uncertainty reporting. These are **recommended extensions**, not proof that TRM4's original physical laws are incorrect. Formal performance acceptance still requires camera-specific measurements, documented uncertainty, and task-based validation.